Rebuttal for ICML 2026

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March 30, 2026

Contents

1 Reviewer TsxC

1.1 Q1 - Can the authors please justify the motivation of their approach in another way beyond the analysis presented in the paper? Less variance in the input by padding with a constant value, resulting in less variance in the activations, is a given and does not directly explain why this works well in one time-series data domain but not others.

We thank the reviewer for this important point. We agree that reduced variance from constant-valued framing, by itself, is not a sufficient domain-level motivation for RIP. We will revise the paper to make it clearer that variance suppression is a mechanistic consequence of the method, not its primary conceptual motivation. Our intended motivation is that RIP changes the label-preserving training distribution in a way that makes narrow or unstable cues less useful. Specifically, RIP augments each training example with structured constant windows carrying the same label, without modifying the internal dynamics of the original sensor window. This is different from standard padding, masking, or naive duplication: padding introduces boundary artifacts, masking alters local dynamics, and duplication increases exposure without constraining how context is used; RIP, instead, introduces structured, non-informative temporal context explicitly designed to be ignored. In this sense, RIP acts less as a simple variance-reduction device and more as a shortcut-reduction regularizer. Because the same class is observed with both the original dynamic window and multiple structured neutral variants, the classifier is discouraged from relying on fragile, generator-specific, or context-specific cues that do not remain predictive across those variants. This interpretation is also consistent with our ablations: naive duplication does not reproduce the effect, randomized replacements for the constants degrade performance, and the invariant frame's ordered structure is essential. Intuitively, this is related to shortcut learning in other domains: models can exploit superficial cues that are predictive during training but not truly class-defining. RIP is designed to counter this behavior by adding neutral structured context that carries the same label, making such shortcuts less useful and encouraging reliance on more robust class-relevant structure. We believe this mechanism is especially well aligned with wearable HAR. In our setting, models trained on synthetic sensor data can overfit spurious temporal context or generator-specific artifacts, and RIP was designed precisely to enforce temporal-context invariance in that synthetic-shift regime. Empirically, the gains are strongest for deep representation-learning models, RIP systematically reduces the synthetic-to-real gap, and it improves LOSO robustness, which is particularly relevant in HAR due to inter-subject variability. By contrast, we do not expect this inductive bias to transfer uniformly to all time-series domains. As already noted in the paper, RIP is most effective in structured, repetitive sensor settings, while gains beyond wearable sensing are modest or inconsistent. Our interpretation is that this occurs because the usefulness of RIP depends on whether label-preserving neutral contextual variants are meaningful for the task. In wearable HAR, this assumption is often reasonable; in other time-series problems, predictive information may depend more strongly on absolute level, trend, regime, or broader context, making such invariance less naturally aligned with the target function. We will revise the manuscript accordingly to distinguish more clearly between:

- (1) the conceptual motivation of RIP as structured context-invariance / shortcut reduction under synthetic shift.
- (2) the mechanistic explanation based on variance propagation that helps explain why this regularization stabilizes learning once that inductive bias is introduced.

1.2 Q2 - Could it be the case that there is something specific about the HAR data domain that makes it amenable to this kind of data augmentation? It's surprising that the improvements do not translate as well to other data domains. It would be great if the authors can discuss this phenomenon.

We thank the reviewer for this insightful question. We agree that our results suggest that the wearable HAR setting is particularly amenable to RIP. Our current interpretation is that RIP is most effective when the task admits a meaningful notion of non-informative temporal context, i.e., when part of the temporal input can be modified in a structured way without changing class identity.

We do not claim that HAR is uniquely special, but rather that RIP appears better aligned with domains in which short, segmented windows admit meaningful, label-preserving neutral variants. Wearable HAR is one such case: activity recognition is typically window-based and must already tolerate segmentation choices, sensor placement effects, and subject variability. In this setting, structured neutral variants are more plausibly compatible with the original label.

By contrast, outside HAR, time-series benchmarks span highly heterogeneous domains, including ECG, electric devices, spectrographs, and image-outline series. Many of these tasks depend more strongly on trend, rhythm, morphology, seasonality, exogenous variables, or broader temporal context. As a result, there is no single shared temporal structure for which one augmentation should be expected to transfer uniformly.

Our interpretation, therefore, is that RIP helps most when structured neutral variants remain semantically compatible with the original label, which seems more common in repetitive sensor settings than in domains where prediction depends more strongly on wider temporal context. We will revise the paper to make this task-alignment hypothesis more explicit: RIP appears best suited to structured, repetitive sensor settings, while its transfer to broader time-series domains is naturally more limited.

1.3 Q3 - Soundness - This paper does not make unsubstantiated claims. The theoretical analysis of the approach is a bit unclear, because the claim is essentially that constant padding of the input reduces variance, which is almost a given. Overall given that the performance of the approach is surprisingly strong given its simplicity, it is important to release the code along with the paper so that these claims can be verified independently.

We thank the reviewer for this comment. We would like to clarify an important point: RIP does not use padding. In padding, a single sample is extended by inserting values at its boundaries, which changes the temporal structure of the original input. This is not what we do. In RIP, the original sample remains unchanged, and we instead construct additional constant-valued samples that are assigned the same label under a structured training scheme. Our theoretical analysis, simulations, and Figure 1 are all based on this setting of constant-valued samples, not padded samples. We agree that this distinction may not have been sufficiently clear in the current manuscript. To address this, we added a new explanatory figure (see Figure 1) and plan to include it in the main paper so that the difference between padding and RIP is explicit to all readers.

Regarding the theory, our intention is not to claim that variance reduction is the primary motivation of the method. Rather, as discussed above, we view it as a mechanistic consequence of training with structured label-preserving neutral variants. The purpose of the theoretical analysis is to study what happens deeper in the model when constant-valued samples with the same label as the original data are introduced, as prescribed by our training construction. In that setting, variance reduction naturally emerges, but we agree that it should be presented as a consequence of the mechanism rather than as its main conceptual justification. We also agree that reproducibility is especially important given the approach's simplicity and unexpectedly strong performance. We therefore intend to release the code upon acceptance so that the community can independently verify the claims and faithfully reproduce the method.

1.4 Q4 - Significance - The performance improvement is very high specifically for the human action recognition scenario. The performance improvements are strong on synthetic and real data. In supplement G, the improvements are marginal if they are present at all. This calls into question the broader significance of this approach, indicating that the theoretical motivation is also mostly valid for one data domain.

We thank the reviewer for this thoughtful comment. We agree that the strongest and most consistent gains are observed in the wearable HAR setting, especially under synthetic-to-real transfer, and that the broader significance of the method should be stated carefully. Our intention is not to claim that RIP is universally strong across all time-series domains. Rather, we view its main significance as introducing a simple and architecture-agnostic temporal context-invariance regularization principle that is particularly well aligned with structured wearable sensing under synthetic shift. In that sense, the non-HAR experiments in Appendix G should be interpreted as preliminary scope tests, not as evidence of uniform gains across all domains. As already noted in the paper, results beyond wearable sensing are modest or inconsistent, but they also rarely degrade substantially. We will revise the paper to make this framing more explicit and avoid overstating broader generality. At the same time, we believe the contribution remains significant for two reasons. First, the improvements in HAR are substantial across both synthetic and real settings, including LOSO robustness and multiple deep architectures, precisely the application regime that motivated RIP. Second, the method is extremely simple, architecture-agnostic, and easy to integrate with standard pipelines, making it a practical regularizer even if its strongest benefits are domain-dependent. We therefore agree with the reviewer that the present evidence most strongly supports RIP as a contribution to structured, repetitive sensor settings, rather than as a universally effective augmentation for arbitrary time-series data. We will revise the manuscript accordingly, positioning the broader-domain results as exploratory and the main significance as a strong, well-supported advance in wearable HAR under synthetic shift.

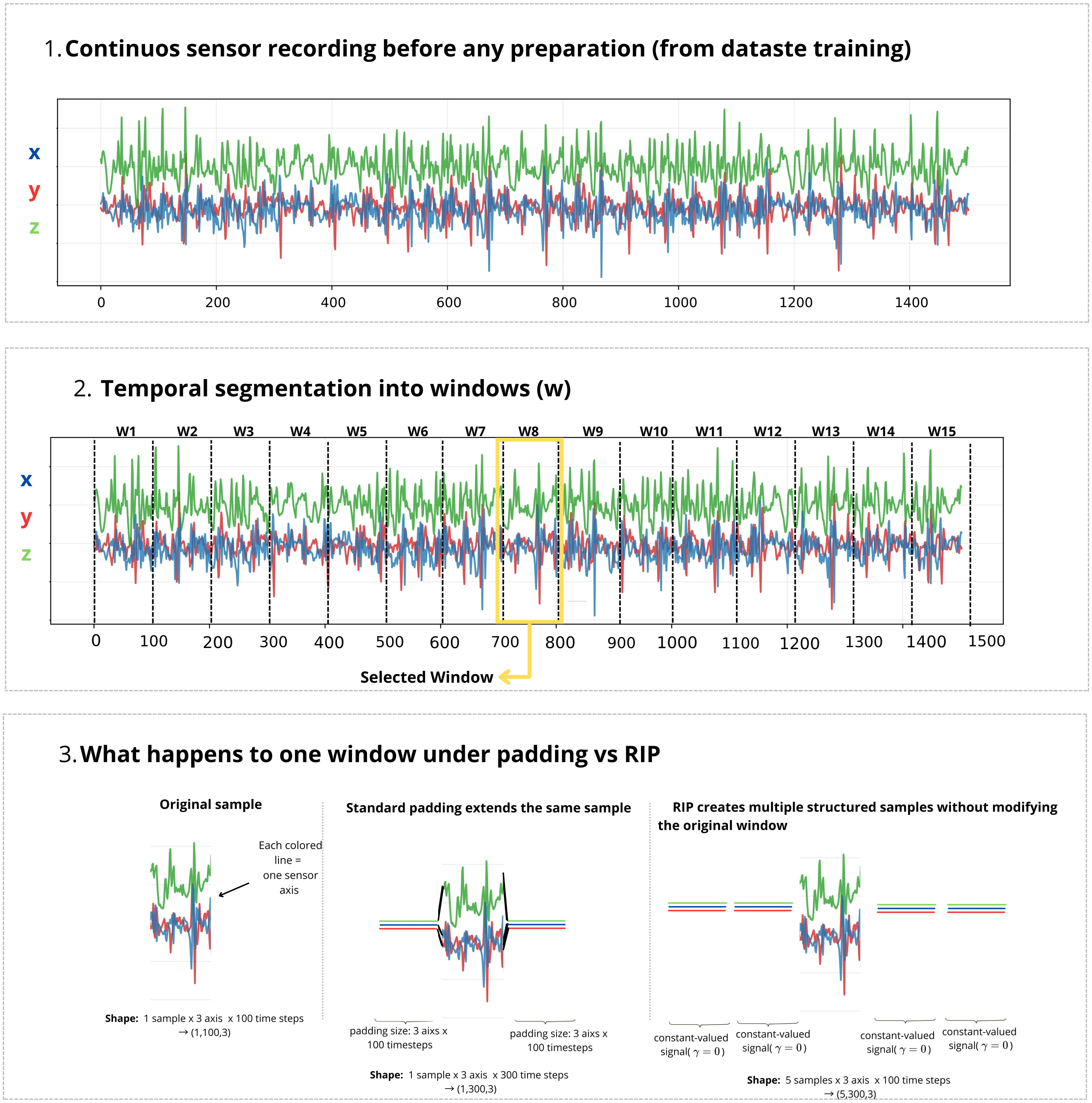

Figure 1: Example of the RIP.

2 Reviewer cc6e

2.1 Q1 - The effectiveness of RIP appears highly specialized for structured, repetitive wearable sensor patterns. Preliminary experiments on general time-series or tabular data showed modest or inconsistent results, raising questions about its universal applicability as a time-series regularizer.

We thank the reviewer for this comment. We would like to clarify that our contribution is not the claim that RIP is a universally effective regularizer for arbitrary time-series data. Rather, the paper makes three more specific contributions. First, we introduce RIP as a simple, architecture-agnostic training mechanism that constructs structured, label-preserving neutral variants, without modifying the original sample or the model architecture. This is a distinct form of regularization that is different from padding, masking, or naive duplication. Second, we show that this mechanism is particularly effective in the setting that motivated the work: wearable HAR under synthetic shift, where it yields strong gains on both synthetic and real data, across multiple architectures and evaluation settings. Third, we provide both ablative and theoretical analyses to understand why the method works: the gains are not recovered by naive duplication, the ordered invariant structure is essential, and the variance analysis is intended as a mechanistic consequence of this construction rather than as a universal explanation across all domains. We therefore agree that the current evidence does not support a universal regularizer claim across arbitrary time-series domains. However, we believe the contribution remains significant because the method is strongly validated in its primary target setting, where the improvements are large, consistent, and practically relevant. We will revise the manuscript to make this scope more explicit and avoid any unintended impression of universal applicability.

2.2 Q2 - Hyperparameter analysis reveals that the duplication factor i accounts for up to 93% of the variance in performance. This suggests the method's success is fragilely dependent on a single parameter that must be tuned specifically for each dataset.

We thank the reviewer for this observation. We agree that performance depends on the duplication factor iii, and our hyperparameter analysis indeed shows that it is the dominant factor. However, we do not view this as evidence of unusual fragility. In practice, most learning-based methods require some degree of dataset-specific tuning, especially when datasets differ substantially in scale, complexity, class structure, and sample availability. In our view, the more important takeaway is positive: although RIP introduces two hyperparameters, the analysis shows that only one of them accounts for most of the performance variation, substantially simplifying practical use. Rather than indicating an overly brittle method, this suggests that RIP has a relatively low-dimensional tuning surface with one main control knob. Moreover, this parameter is interpretable: it directly controls the strength of the structured invariant exposure introduced during training. We also note that its selection can be automated straightforwardly (e.g., with standard hyperparameter search).

2.3 Q3 - The main mechanism of RIP is framing the signal to force focus on the center. However, the paper lacks a detailed analysis of how the model handles the outer-window information.

We thank the reviewer for this comment. We would like to clarify that RIP should not be interpreted as a padding-like mechanism that "forces the model to focus on the center." Padding modifies a sample by extending it with boundary values, thereby changing its temporal structure. This is not what RIP does. In RIP, the original sample remains unchanged; instead, we introduce additional constant-valued contextual samples under the same label through a structured training scheme. With this clarification, we believe that the "outer-window information" mentioned by the reviewer corresponds to these constant-valued contextual samples. This aspect is already analyzed in the paper. Our theoretical investigation was designed precisely to study how the model behaves when trained with such samples, including their effect on deeper layers in both recurrent and transformer-based settings. In addition, we analyze how the inclusion of constant-valued samples changes the effective training distribution seen by the model. Therefore, the paper does not ignore the outer context introduced by RIP. On the contrary, this structured outer context is exactly the object of our theoretical analysis. We agree, however, that this may not have been sufficiently explicit, and we will make it clearer that RIP is about structured context invariance through additional constant-valued samples, rather than padding or center-focused masking.

2.4 Q4 - By framing the dynamic signal within constant windows to enforce focus on the center, is there a risk that the model learns to ignore critical information occurring at the beginning or end of a sensor window?

We thank the reviewer for this question. We believe this concern would apply more directly to a method that modifies, masks, or suppresses parts of the original sensor window. RIP does not do that. The original window is kept unchanged; instead, the method augments training with additional constant-valued samples under the same label. Under this formulation, the relevant question is not whether RIP causes the model to ignore the beginning or end of the original window, but whether adding such label-preserving constant samples leads to loss of class-relevant information. Empirically, our results suggest the opposite: if RIP were systematically suppressing useful class information, we would expect weaker discrimination and poorer generalization. Instead, we observe improved synthetic-to-real transfer and higher overall performance, indicating that the method is not discarding informative structure but rather reducing reliance on unstable or non-robust cues. Our interpretation is therefore that RIP changes the effective training statistics so that constant patterns become less useful for prediction, making variable class-relevant structure comparatively more salient to the model. In this sense, RIP is not designed to suppress boundary information within the original window, but to reduce the predictive value of neutral context introduced through additional constant-valued samples.

2.5 Q5 - Could the gains observed be largely attributed to a simple center-focusing bias? Have you compared RIP against a baseline where models are simply trained on shorter, cropped windows without the constant padding?

We thank the reviewer for this question. We would first like to clarify that RIP does not use padding, and its effect should not be understood as a simple "center-focusing bias." Padding or cropping directly modifies the original sample by extending or shortening it. RIP does neither: the original window is kept unchanged, and the method instead introduces additional constant-valued samples with the same label under a structured training scheme. To make this distinction clearer, we will include an additional explanatory figure in the paper (see Figure 2).

Under this clarification, we do not believe that shorter cropped windows are an equivalent baseline. Cropping removes information from the original signal; RIP does not. RIP preserves the full window and instead changes the training distribution by adding structured constant-valued samples with the same label. These are fundamentally different interventions.

The relevant question, therefore, is not whether "less signal" would work, but whether the gains come from the specific RIP construction rather than from generic regularization or from simply adding more data. Our results support this distinction in two ways. First, naive duplication does not recover the same effect, showing that the gains are not explained by sample count alone. Second, we also compare RIP against standard regularization and data-driven robustness methods, and RIP remains competitive or superior in the target HAR setting. Taken together, these results indicate that the benefit is tied to the specific structured RIP mechanism, not to a generic center-focused bias, a generic more-data effect, or a generic regularization effect.

2.6 Q6 - Your results in Table 2 (TRTR) show significant improvements even when training exclusively on real-world data. Given this, is it accurate to frame RIP primarily as a solution for "Synthetic Shift"? Wouldn't it be more precise to describe RIP as a general temporal regularizer that addresses temporal overfitting inherent in deep HAR models, regardless of the data source?

We thank the reviewer for this important observation. We agree that the gains in Table 2 (TRTR) indicate that RIP should not be viewed only as a mechanism for synthetic-to-real transfer. While the method was originally motivated by the synthetic shift problem in TSTR—namely, improving the utility of synthetic data without changing the model architecture or relying on increasingly complex generative pipelines—our results show that its effect extends beyond that setting.

In this sense, we believe it is more accurate to describe RIP as a temporal regularization mechanism whose benefits are especially pronounced under synthetic shift, but are not limited to it. The strong TSTR gains motivated the method and remain central to the paper, since the synthetic shift is the most challenging and practically relevant setting we target. At the same time, the TRTR results suggest that RIP also mitigates a broader form of temporal overfitting in deep HAR models, even when training only on real data. We therefore view the most precise framing as follows: RIP is a general temporal regularizer for wearable HAR, with particularly strong impact in synthetic-shift scenarios. This is also why we continue to emphasize the synthetic shift in the title and positioning: it is the original motivation and the setting where the practical impact is largest, even though the method is not restricted to that case.

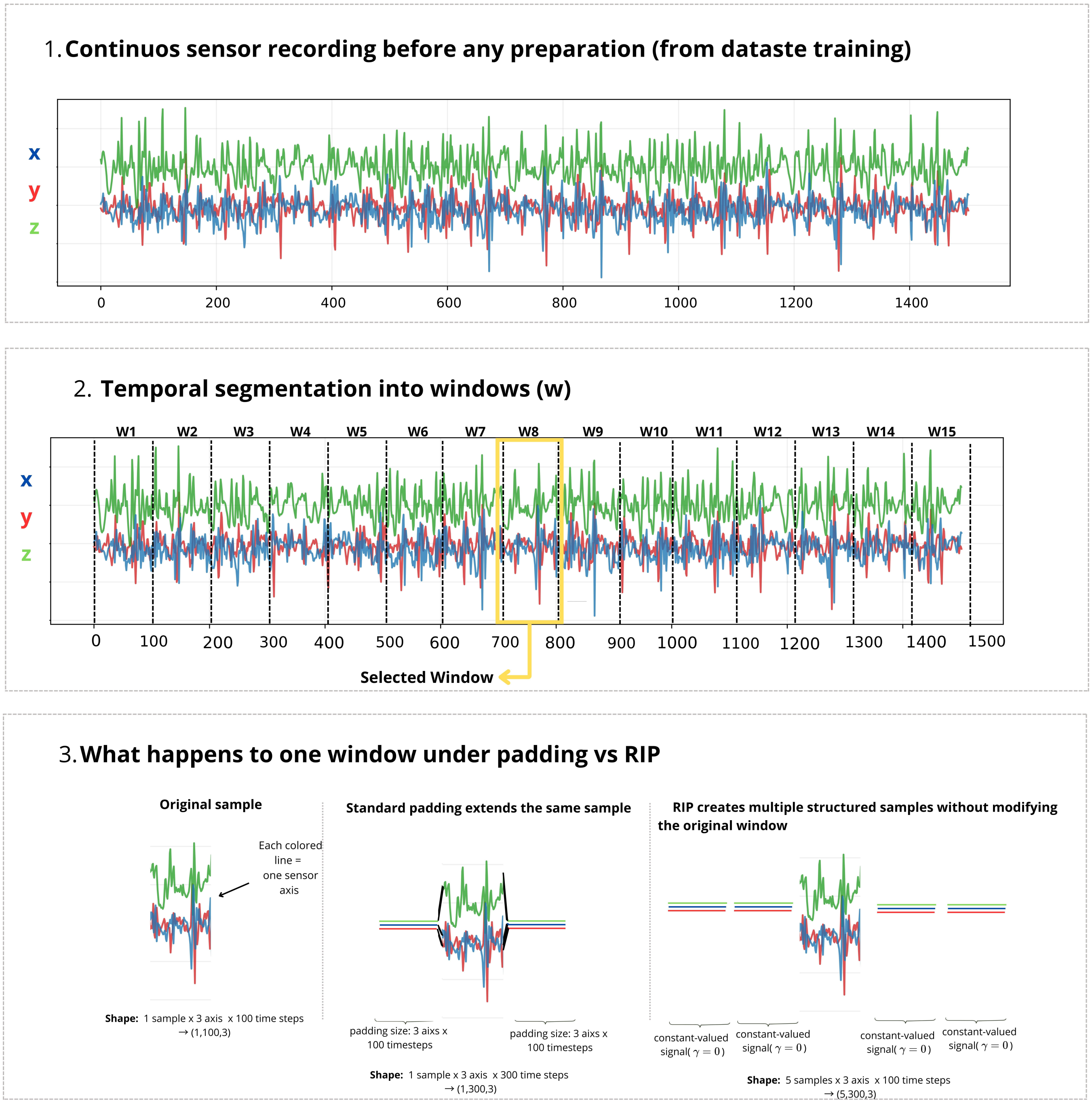

Figure 2: Example of the RIP.

3 Reviewer LLQK

3.1 Q1 - The selection of hyperparameters lacks universal guidance: the experimental distribution of the optimal replication factor is quite scattered, indicating that the regularization strength needs to be adjusted according to the characteristics of the dataset. Although a search method is provided, the lack of parameter selection principles related to data attributes (such as sequence length and signal complexity) increases the tuning cost in practical applications.

We agree that performance is primarily controlled by the duplication factor iii, which is the dominant hyperparameter. However, we do not interpret this as a sign of fragility. On the contrary, this indicates that RIP has a low-dimensional and interpretable tuning space, with essentially one main control knob. The parameter iii directly controls the strength of invariant exposure during training and behaves similarly to a regularization strength parameter. It can be tuned using standard validation procedures, and its selection can be automated with common search strategies. Thus, rather than increasing complexity, RIP provides a simple, practical tuning mechanism.

3.2 Q2 - The experiments only used TLC-GAN to generate synthetic data, failing to explore whether RIP is equally effective on other generators (such as Diffusion models and VAEs).

RIP is generator-agnostic by design, operating purely at the data level and not relying on generator-specific properties. The consistent improvements across datasets and architectures suggest that the effect is not tied to a particular generator artifact. To further address this concern, we evaluated RIP against an alternative synthetic generation method using large language models [1]. The results show that RIP still provides substantial improvements over the synthetic baseline, the synthetic F1 baseline was 16.85+- 2.53, while RIP achieved 26.72+-8.01. This is consistent with our claim: RIP helps mitigate synthetic shift, but does not eliminate it completely. We acknowledge that evaluating more generators is valuable future work and will clarify this limitation in the manuscript. The Figure `https://anonymous.4open.science/r/RIP-5180/gemini_data.png` shows the complete result for this experiment.
[1] Souza et al., "Evaluating the Usefulness of Large Language Models for Synthetic Samples Generation via Few-shot Learning," LatinX in AI @ NeurIPS 2024.

3.3 Q3 - RIP is formally similar to data augmentation methods such as Cutout and Masking (all introduce non-informative regions). Although the paper makes a distinction in Section 2.1, the emphasis on the fundamental difference of "temporal context invariance" is insufficient, which may be misinterpreted as another augmentation technique rather than a new regularization paradigm.

We appreciate this important question. While RIP may superficially resemble augmentation methods such as masking or cutout, we emphasize that its role is fundamentally different, particularly in the temporal domain. In computer vision, non-informative regions (e.g., backgrounds) naturally exist and vary across samples, enabling domain randomization to encourage invariance to such factors. In contrast, time-series data—especially in HAR—does not exhibit an explicit notion of "background": the signal is fully occupied by temporal dynamics, and non-informative context is not readily separable. RIP can therefore be understood as introducing an explicit analogue of "background" into time-series data. The constant-valued windows serve as a controlled, non-informative temporal context that does not naturally occur in the data but is necessary to enforce invariance to spurious temporal context. In Cutout/Masking, the original sample itself is altered by suppressing part of its content. In contrast, RIP does not modify the original sample at all, nor does it mix it with another sample. Instead, RIP changes the training distribution by introducing additional constant-valued label-preserving samples under a structured training scheme. Thus, the key distinction is not only what is inserted, but what is being regularized. For this reason, we do not view RIP as simply another augmentation technique. Standard masking-style methods promote robustness to local corruption of the input. RIP, instead, is designed to enforce invariance to non-informative temporal context through a structured, label-preserving data construction, without changing the model architecture or the original signal itself. This is why we frame it as a data-centric regularization mechanism rather than as a conventional augmentation method. This differs from standard augmentations in two key ways:

- (i) augmentations (e.g., masking, jitter) perturb or remove parts of the signal, often altering local dynamics;
- (ii) RIP preserves the original signal intact and instead modifies the surrounding temporal context in a structured and deterministic way.

As a result, RIP does not increase data diversity but enforces a constraint on how the model uses temporal context, which we formalize as temporal context invariance (Section 2.1) and analyze via variance suppression (Section 3). We will strengthen this intuition in the revised manuscript to clarify that RIP is not a conventional augmentation technique, but a temporal analogue of domain randomization that introduces a missing inductive bias in time-series learning.

3.4 Q4 - The core mechanism of RIP is to reduce the variance of the hidden states by inserting a constant window, thus achieving implicit regularization. Could this variance suppression effect be simply attributed to "the training samples becoming longer" or "the effective batch size changing"? Although the paper compares its performance with other data-driven regularization methods in Table 5, are there any controlled experiments (such as directly replicating the original window or inserting a random noise window) to demonstrate that RIP's unique effect stems from the "constant value" design choice, rather than simply "providing more context"?

We thank the reviewer for this insightful question. We first clarify that the interpretation in terms of "making samples longer" does not apply to RIP. RIP does not extend or pad the original window; the sample remains unchanged. Instead, RIP changes the construction of training batches by introducing additional constant-valued samples with the same label in a fixed, ordered structure around the real sample. The mechanism, therefore, does not rely on altering the temporal extent of the original data but on exposing the model to a structured label-preserving context.

Under this formulation, the relevant question is not whether "longer samples help," but whether the gains can be explained by simply adding more samples. Our ablation study (Section 7) directly addresses this: (i) naive duplication of the original window does not recover RIP's gains; (ii) replacing constant-valued windows with random noise degrades performance; and (iii) unordered / non-structured context also leads to clear drops. These controls show that the effect is tied to the specific constant-valued, ordered construction of RIP, rather than to a generic more-data, more-context, or batch-size effect.

We also compare RIP against standard data-driven regularization methods in Table 5, where RIP remains clearly stronger in the target HAR setting. This further supports that the benefit is not reducible to generic augmentation or generic regularization alone. Finally, from the theoretical perspective of Section 3, the variance reduction arises because RIP replaces multiple stochastic temporal contributions with a deterministic invariant component. This effect follows from the specific invariant-pattern design, not from simply increasing sequence length or effective batch size.

3.5 Q5 - Have you considered adding new baselines from the last two years (2024&2025)?

We thank the reviewer for this suggestion. Our current evaluation focuses on widely adopted baselines in HAR and time-series classification to isolate the contribution of RIP. We also emphasize that RIP is complementary to these approaches, as it operates at the data level and it is not architecture-dependent, but it can be combined with modern architectures and training paradigms.

4 Reviewer Ssh3

4.1 Q1 - Soundness: However, significant concerns remain regarding the transparency of the results and implementation. Several reported gains in the TSTR protocol are extraordinarily large, such as the WHARF F1-score jumping from a baseline of 6.1% to 87.3%. Such atypical improvements warrant a closer diagnostic to ensure no evaluation leakage or window overlap occurred. Additionally, there is a clear discrepancy between the described 4i + 1 data expansion and the claim that training time with only i=16 "approximately doubles". For i=16, the data volume per sample increases 65-fold, making the reported computational overhead appear incongruent with standard GPU scaling. Finally, while the manuscript emphasizes that the "order is crucial" in RIP construction, it fails to explain how this order is maintained in pipelines that utilize stochastic data shuffling.

We thank the reviewer for pointing this out. We apologize for the confusion. After revisiting our timing and memory measurements, we realized that the earlier comparison had been performed with different batch settings, which made the reported overhead misleading. We have now repeated the measurements on WISDM, our largest dataset, using the same batch size and the largest tested configuration, $i = 16$. With this corrected setup, the computational increase is indeed substantial and consistent with the actual RIP data expansion. We also identified that the manuscript's notation was not sufficiently explicit. The expansion factor should be described at the dataset level as: expanded dataset size = $(2i + 1) \times N$ where N is the number of original windows. In the corrected WISDM experiment, the original training set had shape (20846, 100, 3), while RIP with $i = 16$ had shape (687918, 100, 3), which corresponds exactly to a 33x expansion, since $2 \times 16 + 1 = 33$. This confusion arose because we used the notation $k = 2i$ to describe the algorithm. We will correct this notation in the paper to avoid the misleading impression that the increase was only approximate or minor. Under the corrected measurement, the mean training time increased from 42.87 s to 1683.64 s, and peak GPU memory increased from 305.09 MB to 3417.23 MB. These revised numbers are therefore consistent with the actual data expansion used in training.

Regarding the reviewer's concern about order under stochastic shuffling, the crucial order in RIP is the order within each constructed sample, not the order in which samples are presented across batches. Each RIP sample is created with a fixed internal structure, in which the constant windows and the real window maintain their relative positions. Standard dataloader shuffling only permutes the order of samples across batches; it does not alter the internal order of windows inside each RIP sample. Therefore, stochastic shuffling is fully compatible with RIP while preserving the method's ordered structure.

4.2 Q2 - Presentation: However, the manuscript lacks sufficient implementation detail to fully support its "architecture-agnostic" claims across diverse model families. While the authors define augmented samples as sequences of windows, they do not explicitly clarify how this expanded temporal input is ingested by non-recurrent classifiers like CNNs or Random Forest without modifying input layers or processing logic.

We thank the reviewer for this comment. We would like to clarify what we mean by architecture-agnostic in the paper. Our claim is not that RIP requires no implementation detail for every possible model family, nor that we exhaustively validate it on all architectures. Rather, the claim is that RIP operates at the data-construction level and therefore does not require modifying the model architecture or loss. This is the sense in which we use the term architecture-agnostic in the manuscript.

Concretely, RIP does not alter the input layer or the classifier's processing logic. The method constructs additional constant-valued samples with the same label under a structured training scheme, while keeping the original sample unchanged. These constructed samples are then passed to each classifier in the same input format that the classifier already expects.

For non-recurrent models, this does not require expanding the architecture to ingest a new kind of temporal object. The classifier simply receives the RIP-constructed samples in the same representation it normally uses: sequence models process them as ordinary input windows, and feature-based models extract their usual statistics/subsequences from those windows before classification. In that sense, RIP is architecture-agnostic because the intervention is upstream of the classifier, at the level of the training data, rather than inside the model.

We agree, however, that this point may benefit from a more explicit implementation-level explanation in the paper. The intended claim is therefore practical rather than universal: RIP is model-agnostic across the families we evaluate, because it can be applied without altering their architectures or losses, even though the downstream classifier may be recurrent, convolutional, transformer-based, or feature-based.

4.3 Q4 - Originality: However, the originality is tempered by a limited engagement with existing time-series invariance literature. The manuscript provides sparse comparisons to established Self-Supervised Learning (SSL) or Invariant Risk Minimization (IRM) frameworks that are currently the benchmarks for domain generalization in time-series analysis. The discussion of standard augmentations like time-warping or SpecAugment-style masking is also relatively brief.

We thank the reviewer for this thoughtful comment. We agree that the discussion of SSL/IRM and related invariance literature can be made more explicit. However, the paper's originality claim is not that RIP is another generic domain-generalization framework, but that it introduces temporal context invariance as a data-centric regularization principle for time-series learning under synthetic shift, instantiated without changing architectures or losses. The manuscript also engages with standard time-series augmentations: Section 2.1 explicitly discusses jitter, scaling, time-warping, cropping/permutation, temporal masking/cutout, Mixup/CutMix, and SAM/DRO, and positions RIP against them as targeting a different invariance—namely, invariance to non-informative temporal context. In addition, although we do not present RIP as a full SSL/IRM-style domain-generalization framework, we did include LOSO evaluation precisely as a good-faith test of generalization. This was important to show that the observed gains are not merely due to mixed-subject training data, but remain present even when the model is evaluated on a previously unseen subject. In this sense, the paper addresses generalization concerns in the practical HAR setting, even if its contribution is more targeted than that of the broader SSL/IRM literature. RIP fundamentally differs from both Invariant Risk Minimization (IRM) [1] and classical data augmentation approaches in where and how invariance is enforced. IRM explicitly requires multiple environments and learns a representation such that the same optimal classifier holds across them, enforcing invariance through a constraint on the predictor itself; in contrast, RIP does not rely on explicit environment partitioning or bi-level optimization, but instead induces invariance implicitly through inserting invariant windows within the data (e.g., replication or masking patterns) that simulate invariance conditions during training. Compared to standard augmentation methods (e.g., masking or Cutout), which inject random or task-agnostic noise, RIP is not designed to increase data diversity, but to enforce a specific invariance assumption about temporal/contextual structure, making the perturbation semantically aligned with the underlying signal rather than arbitrary. In short, while IRM enforces invariance at the representation–predictor level and augmentation operates at the input level without guarantees, RIP occupies a middle ground: it operates at the input level but is principled and targeted, aiming to bias the model toward invariant signal components without requiring explicit environment supervision or causal assumptions. Our claim is therefore not universal superiority over all invariance frameworks, but a distinct and practically effective data-centric regularization mechanism for structured wearable sensing. [1] Arjovsky, Martin, et al. "Invariant risk minimization." arXiv preprint arXiv:1907.02893 (2019).

4.4 Q5 - Key Questions For Authors: 1.Several reported TSTR gains are extraordinarily large (e.g., WHARF F1: 6.1 $\rightarrow$ 87.3; WISDM near 99.7%), which is atypical and raises concerns about potential evaluation leakage, window overlap issues, or overly easy test protocols; stronger controls and diagnostics are needed.

We thank the reviewer for this thoughtful comment. We performed additional sanity checks:

- (i) strict separation between synthetic training data and real test data under the TSTR protocol, ensuring no overlap at the subject or window level;
- (ii) verification that sliding windows do not cross train/test boundaries;
- (iii) evaluation under LOSO (Leave-One-Subject-Out), which also shows consistent improvements (Table 4 in the manuscript).

These controls confirm that the observed gains are not due to leakage but reflect improved generalization under a synthetic-to-real shift. We will include the additional diagnostics (e.g., confusion matrices and subject-wise results) in the revised version to further support this claim.

4.5 Q6 - Key Questions For Authors: 2.Several TSTR results are near-perfect (e.g., WISDM Accuracy/F1 $\approx$ 99–100%). Can you include diagnostic analyses (e.g., per-class confusion, subject-wise breakdowns, and sanity checks on label distribution and window overlap) to rule out artifacts?

We thank the reviewer for this important question. In response, we conducted additional sanity checks aimed at checking data leakage and related artifacts, including inspection of the train/test pipeline, verification of label distributions, and checks to ensure that training samples were not inadvertently included in the test set.

During this investigation, we identified an implementation issue affecting only the TSTR results of the Dclassifier. Specifically, due to a typo in the condition used to select the synthetic training split, that experiment was executed with real + RIP data instead of the intended synthetic + RIP data. We emphasize that this issue is restricted to this specific classifier/configuration; we did not find evidence of the same problem in the other classifiers or in the TRTR experiments.

After identifying the issue, we re-ran the affected experiments and updated the corresponding TSTR results for DClassifier in Table 1. Importantly, the corrected results reduce the previously reported magnitude of the gain, but they do not invalidate RIP. The corrected DClassifier TSTR results still show consistent improvements over the synthetic baseline across all five datasets (see Figure 3):

We are also including confusion matrices for WISDM in both TRTR and TSTR (in the end of this answer on figures 4, 5), with and without RIP, as additional diagnostics. These support the same interpretation: in TRTR, RIP sharpens the diagonal and reduces major confusions (e.g., for classes 0, 4, and 5), and in TSTR, although the synthetic-real gap remains substantial, RIP still improves class discrimination relative to the synthetic baseline. In particular, the corrected WISDM TSTR result increases from 40.11 ± 3.92 F1 to 48.14 ± 6.64 with RIP.

Model	Dataset	RIP Config.	Accuracy (%)		F1-score (%)	
		(γ, i)	Baseline	RIP	Baseline	RIP
DClassifier	MHEALTH	(5, 16)	55.79±2.44	**59.64±0.37**	52.56±2.33	**54.82±0.34**
	MHAD1	(5, 16)	29.60±1.40	**32.78±1.46**	27.07±1.20	**31.02±1.62**
	MHAD2	(5, 16)	46.97±3.70	**54.90±3.76**	43.52±3.78	**49.56±3.91**
	WHARF	(5, 5)	15.46±3.04	**20.17 ±1.0**	6.14±1.78	**8.21 ±1.10**
	WISDM	(0, 16)	43.23±4.40	**56.61 ±0.11**	40.11±3.92	**48.14 ± 6.64**
	MHEALTH	(0, 1)	[illegible]	[illegible]	[illegible]	[illegible]

Figure 3: New results for DClassifier.

Although this correction changes the magnitude of the effect for the DClassifier TSTR case, we believe it ultimately strengthens the paper by ensuring that the reported conclusions are based on the correct experimental pipeline and transparent diagnostics.

4.6 Q7 - Key Questions For Authors: 3.Please include stronger baselines: standard time-series augmentations (time-warp, jitter, scaling, channel dropout, time masking/cutout), and SSL pretraining (TS-TCC/SimCLR variants), as well as modern domain adaptation baselines. How does RIP compare under these stronger controls?

We thank the reviewer for this suggestion. We agree that stronger controls are important. At the same time, we would like to clarify the paper's intended scope. Our goal is not to present RIP as a universal benchmark against every possible time-series augmentation, SSL pretraining framework, or domain-adaptation pipeline. Rather, the contribution is a simple data-centric regularization mechanism that can be applied without modifying the model architecture or training objective, and whose main target setting is wearable HAR.

For this reason, we focused our comparisons on representative and practically established data-centric baselines already used in HAR and closely related robustness settings (Table 5 in the main paper). In particular, we compared RIP against Mixup and CutMix, which are strong and widely used augmentation baselines in this family. On MHAD2 with the Dclassifier, RIP clearly outperformed both (see Figure 6):

These results support that RIP's gains are not explained by a generic augmentation effect. Even against strong data-mixing baselines, RIP remains clearly superior in the target HAR setting. We did not include a broader suite of traditional augmentations, such as time-warping, jitter, scaling, or channel dropout, because these were not the main focus of the paper, and our intention was not to exhaust all augmentation variants. More importantly, our current baselines already test the more relevant alternative explanation: whether RIP can be reduced to a generic data-centric regularization/augmentation effect. The evidence above suggests that it cannot.

Regarding SSL pretraining and modern domain adaptation, we would also like to clarify the scope. RIP is not proposed as a full SSL framework or as a dedicated domain-adaptation method. Likewise, our use of LOSO was not intended as a full domain-generalization benchmark, but rather as a good-faith robustness check to show that RIP's gains do not depend on mixed-subject training and remain present even when evaluating on unseen subjects. In other words, LOSO was included to test broader robustness in HAR, not to reposition the paper as a domain-adaptation

WISDM - TRTR						
	pred_0	pred_1	pred_2	pred_3	pred_4	pred_5
true_0	11715	538	55	63	3023	2846
true_1	255	64319	0	0	295	351
true_2	13	3	11185	330	76	3
true_3	17	0	73	9183	57	0
true_4	1860	1364	22	101	16404	2899
true_5	1315	43	49	13	1505	78345
WISDM - TRTR + RIP (gamma = -1, i = 16)						
	pred_0	pred_1	pred_2	pred_3	pred_4	pred_5
true_0	16691	111	1	11	919	507
true_1	109	64794	0	0	168	149
true_2	31	1	11449	88	33	8
true_3	38	0	10	9266	15	1
true_4	667	272	1	14	21262	434
true_5	329	31	14	4	326	80566

Figure 4: Confusion-matrix for TRTR.

work.

We therefore believe the most accurate framing is: RIP is a targeted, practically effective regularization mechanism for wearable HAR, and the current comparisons already show that its gains are not reducible to standard data-mixing baselines or generic augmentation alone.

4.7 Q8 - Key Questions For Authors: 4.The computational overhead claim ("i=16 approximately doubles training time") seems inconsistent with the 4i+1 expansion. Could you clarify the data pipeline and provide measured wall-clock training times and memory usage per i and per model?

We answered this question in the Q1 answer.

4.8 Q9 - Key Questions For Authors: 5.Could simpler context manipulations (e.g., zero/constant padding of longer inputs or randomized context lengths) achieve similar effects? Please provide a direct ablation vs padding and vs SpecAugment-style time masking.

We thank the reviewer for this question. We addressed this point in good faith by testing a simpler context manipulation on the MHAD2 dataset (both real and synthetic versions), using the same RevTransformer training setup adopted in our RIP experiments. Specifically, we evaluated a time-masking baseline in which 10 values within each 50-step window were randomly set to zero. This baseline did not provide evidence that a simple masking strategy can explain RIP. Its behavior was considerably less stable across settings. On real MHAD2, time masking performed substantially worse than both the baseline and RIP. On synthetic MHAD2, although its mean F1 was higher, it came

WISDM - TSTR						
	pred_0	pred_1	pred_2	pred_3	pred_4	pred_5
true_0	5175	1520	295	2633	1509	6808
true_1	3570	50135	895	129	743	9548
true_2	1022	1364	8417	420	134	243
true_3	2367	147	3975	893	252	1696
true_4	6580	2657	523	3675	1827	7388
true_5	18988	4909	517	5850	6474	44532
WISDM - TSTR + RIP (gamma = -1, i = 16)						
	pred_0	pred_1	pred_2	pred_3	pred_4	pred_5
true_0	5272	1040	296	4251	1298	6083
true_1	3124	50260	327	91	3413	8005
true_2	725	511	8496	674	619	585
true_3	1326	62	46	6962	268	666
true_4	6175	2165	421	5660	1827	6402
true_5	16175	3398	614	9484	6470	45129

Figure 5: Confusion-matrix for TSTR.

with a much larger confidence interval/variance, whereas RIP remained markedly more stable. Figure 7 present this results:

We therefore do not see simple masking as a reliable substitute for RIP. Even when it improves the mean in one setting, it does so with substantially higher variability, whereas RIP is based on a controlled, structured construction that yields more stable behavior. We did not include a padding baseline because padding is not equivalent to RIP. Padding changes the temporal length of the original window, whereas RIP leaves the original window unchanged and instead changes the training-set construction by adding structured constant-valued samples under the same label. In our pipeline, padding would require altering the downstream input specification and would therefore not constitute a fair controlled comparison. In addition, our ablation study already addresses closely related alternatives: naive duplication does not reproduce the effect, random context degrades performance, and structure/order are critical. Taken together, these controls indicate that RIP is not reducible to simple padding, masking, or generic context manipulation; its effect depends on the specific structured constant-valued construction. In addition, we have already addressed this question in our ablation study (Section 7), where we show that:

- naive duplication does not reproduce the effect,
- random context degrades performance,
- structure and ordering are critical.

These results indicate that RIP is not equivalent to simple padding or masking.

MHAD2 -TRTR- Dclassifier		
Approach	Accuracy	F1
Baseline	68.32±1.05	67.58±1.31
RIP (0, 16)	82.12±0.72	81.68±0.74
Mixup	39.98±12.17	30.89±16.12
Cutmix	63.10±1.18	62.03±1.47
MHAD2 -TSTR-Dclassifier		
Baseline	46.97±3.70	43.52±3.78
RIP (5, 16)	54.90± 3.76	49.56± 3.91
Mixup	39.88±6.14	32.33±9.63
Cutmix	45.38±2.83	41.80±4.05

Figure 6: CutMix and MixUp results.

4.9 Q10 - Key Questions For Authors: 6.Have you tested different synthetic generators or real-to-synthetic mismatch regimes to assess robustness of RIP to generator artifacts?

We thank the reviewer for this important question. In the original submission, we did not include multiple synthetic generators because our study was built on TLCG, which had already been evaluated against more traditional synthetic baselines in prior work and showed stronger performance. Our goal in this paper was therefore not to exhaustively benchmark generators, but to investigate whether RIP could improve the usefulness of synthetic data in a strong-generator setting.

That said, to address the reviewer's concern, we additionally evaluated RIP under a different synthetic generation pipeline using the protocol described in: Souza et al., "Evaluating the Usefulness of Large Language Models for Synthetic Samples Generation via Few-shot Learning," LatinX in AI @ NeurIPS 2024.

We extracted both the reported results and the released data from the corresponding repository and evaluated the generated samples with RIP. Figure 8 shows this results.

These results suggest that RIP can still provide gains under a different synthetic-generation regime, substantially improving over the synthetic baseline even when the generator and mismatch characteristics differ from those in our main experiments. At the same time, the absolute performance remains below the real-data baseline, consistent with our general claim: RIP helps mitigate synthetic shift but does not eliminate it completely.

We therefore do not claim exhaustive robustness across all possible generators or mismatch regimes. Rather, this additional experiment provides evidence that RIP is not tied to a single generator, and that its benefits can extend to at least one alternative synthetic pipeline beyond TLCG.

Dataset	Classifier	Approach	F1	Setting
MHAD2	RevTransformer	Baseline	69.03 ± 1.0	-
MHAD2	RevTransformer	RIP	57.47 ± 2.0	(γ = 1, i = 5)
MHAD2	RevTransformer	time masking	41.80±4.05	(v=1, s= 5)
Synthetic MHAD2	RevTransformer	Baseline	44.29 ± 3.0	-
Synthetic MHAD2	RevTransformer	RIP (γ = 1, i = 16)	44.98 ± 3.0	(γ = 1, i = 16)
Synthetic MHAD2	RevTransformer	time masking	43.82 ± 5.15	v=0, s= 10)

Figure 7: New results for time masking.

Approach	ACC	F1
Real baseline	69.55+-2.42	68.53+-2.41
Synthetic Baseline	19.69+-3.39	16.85+- 2.53
RIP(0,16)	32.72+-12.04	26.72+-8.01

Figure 8: New results for LLM time-series generator.